Score-linked practice intelligence from long-form violin training video

Alan N. Pham[1]

[1]AO Labs, Worcester, MA, USA

High-level music training produces dense evidence: daily practice videos, repeated repertoire, changing technical constraints, and periodic performance goals. Most of that evidence is difficult to use because it is stored as unstructured media rather than as a score-linked longitudinal record. We report Curtis, a live AO Labs system that converts messy long-form violin practice videos into structured, evidence-linked practice records using active practice detection, sheet-music-style transcription, score alignment targets, heat maps, repertoire tracking, and passage-level performance observation. The checked live state on 7 May 2026 used the YouTube channel `@nalalan` as the primary source, indexed 209 public videos, marked 104 as practice candidates, marked 84 as long-form candidates, stored 20 media samples, retained 51 reviewed sections, and recorded 80 Curtis-focused findings. The uploaded-video ledger begins at `violin 1`, uploaded on 20 December 2025, and currently counts 40 violin-numbered or date-titled practice logs totaling 158 h 2 min of public session duration; this duration is not counted as active practice time. Active violin practice time is counted only from detected playing windows after media processing. The primary interface has three top-level surfaces: a paper card, analyzed practice days, and repertoire. Each daily record can show uploaded duration, measured active violin time, machine-generated sheet-music-style notation, evidence clips, repeated-fragment heat maps, confirmed or uncertain piece evidence, and specific observed blockers such as uncertain repeated notes, longer pause or restart markers, slow windows, and repeated fragments. Repertoire entries are promoted only from confirmed daily-record evidence and keep possible labels, user-rejected false labels, weak evidence, and readiness scores inferred only from piece identification separate from confirmed repertoire and judged playing quality. The system uses GPT-5.5 for vision and text review and separate audio models for identification and verification, while preserving explicit limits: Curtis admission cannot be predicted from current samples, weak audio/video evidence is marked unjudged, and the system reports practice signals rather than admissions

probability. The contribution is an experimental method for making practice visible, searchable, measurable, and tied to actual musical evidence.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (*1*, *2*). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis addresses this problem as a live software system rather than as a portfolio page (*3*). The system indexes public practice media, separates uploaded video duration from detected active violin practice, records technical observations, and keeps evidence tied to clips and generated notation. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and identify the specific passage-level blockers that can be tested in subsequent recordings.

The core pipeline is video → active practice detection → clips → music transcription → repeat grouping → score matching → heat map → repertoire update → passage-level observations. The output is intentionally small: analyzed practice days and an automatically updated repertoire list, both grounded in clips, notation, active playing time, and confidence limits.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (*4*). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The first production state used the live backend at `https://curtis.aolabs.io/api/curtis/media-status`. The checked state on 7 May 2026 reported the values in Table 1. These are operating facts, not outcome claims.

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal

Table 1. Curtis production state on 7 May 2026.

Signal	Current value
Primary source	YouTube channel `@nalalan`
Inventory	209 indexed videos
Practice candidates	104 videos
Long-form candidates	84 videos
Uploaded-video marker	`violin 1`, uploaded 20 December 2025
Title-confirmed uploaded-video ledger	40 videos, 158 h 2 min uploaded session duration
Active practice time	Counted only from detected violin-playing windows; uploaded duration is reported separately
Media samples	20 captured samples
Reviewed sections	51 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-3-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training anchors	3 source-confirmed practice sources
Score-reference targets	3 configured
Analyzed practice days	40 title-confirmed daily records; 3 confirmed source labels with score snippets
Pitch/rhythm transcription	Filtered monophonic violin-range pYIN configured; per-sample note events and fingerprints stored when media samples exist
Specific observations	Generated only from extracted uncertainty, pause/restart, tempo, and repetition evidence
Score-aligned windows	0; pitch/rhythm-to-score matching is not yet configured
Curtis-level completion	Withheld when the available evidence only identifies repertoire
Current focus	Audition-style setup consistency
Current constraint	Open strings only, mirror-check contact point
Boundary	Curtis admission cannot be predicted from current samples

records because the title contains a practice date and the duration preserves session-level evidence. The uploaded-video ledger starts at the `violin 1` marker and counts only violin-numbered or date-titled practice logs, so unrelated long public videos are retained in inventory without being silently counted as violin practice. This ledger is not the active-practice total. Active practice is measured only after media processing identifies windows where violin playing is actually occurring.

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

3 First readout

The first live readout is experimental but usable. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples, stored section reviews, and title-confirmed practice records. The home screen has three top-level surfaces: a paper card, analyzed practice days, and repertoire. Each analyzed day groups same-day videos, preserves uploaded duration, shows active-playing status, renders generated notation when pitch/rhythm extraction exists, links clips, shows repeated-fragment heat maps, and names confirmed or uncertain piece evidence without promoting guesses. The repertoire surface is evidence-backed: an entry appears only when a daily record carries confirmed source evidence, and each entry keeps practice dates, clips, notation snippets, active time when measured, and current blocker observations. Generic etude/caprice evidence remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from stored review findings, sample indexes, source corrections, public-domain score targets, machine note/rhythm events, heat-map fragments, and the uploaded-video ledger.

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification and evidence quality. YouTube metadata can identify candidate videos, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. The current training ledger separates source-confirmed labels from independent audio-title matches, machine pitch/rhythm windows, and score-aligned windows. It also stores reference targets for the three confirmed source labels across two works so later passes can ask for section-level matching instead of repeating broad title guesses. The transcription path converts sampled media to mono audio, estimates monophonic violin pitch with a pYIN pitch track bounded to the violin range, rejects out-of-range pitch artifacts, requires sustained pitch changes before opening a new note event, estimates tempo, stores compact note-duration notation, and builds pitch-class, interval, rhythm, and contour fingerprints for future matching. A stronger system should align those events against a symbolic or optical score representation, identify movement sections or measure ranges, compare against reference recordings, and then judge articulation, timing, tone, and tempo against the aligned passage.

The broader experiment is whether a practice system can make musical development legible over time. A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

5 Materials and methods

Backend

The backend is a FastAPI service deployed on Railway. It exposes `https://curtis.aolabs.io/api/curtis/media-status` for compact state, `https://curtis.aolabs.io/api/curtis/ops-check` for fuller operational state, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

The default public source is `https://www.youtube.com/@nalalan`. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The practice-hours total uses a stricter title-confirmed ledger from `violin 1` onward: violin-numbered videos and date-titled logs count toward the top-line session duration, while unrelated long videos remain indexed but outside the practice-hours total. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, early setup windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV. Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as anchors and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints, and future samples can receive a nearest-source hint when their extracted notes and rhythms resemble a prior confirmed source. Possible labels and model-only verified titles remain uncertain daily evidence unless a source label is confirmed, so future failures appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. The daily-record builder joins the strict `violin 1` ledger to media samples, machine note/rhythm events, active-time evidence, clips, score packet support, repeated-fragment heat maps, and observation records. Repertoire entries are built only from confirmed daily-record evidence and keep progress percentages unassigned until clips, notation, and score alignment support them.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

6 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The useful claim is therefore limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

References

1. K. A. Ericsson, R. T. Krampe, C. Tesch-Romer, The role of deliberate practice in the acquisition of expert performance. *Psychological Review* **100**, 363–406 (1993).
2. A. Williamon, Ed., *Musical Excellence: Strategies and Techniques to Enhance Performance* (Oxford University Press, Oxford, 2004).
3. A. N. Pham, *Curtis Media Review*, Live software system, curtis.aolabs.io, 2026.
4. A. N. Pham, *AO Progress Ledger*, Live longitudinal state system, progress.aolabs.io, 2026.